



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Business Systems

Academic Year 2024– 2025(Odd Semester)

Degree, Semester & Branch: B.Tech – I Semester CSBS

Course Code & Title: GE3151 & Problem solving and python programming

Name of the Faculty member (s): Mrs.B.Yazhini,AP/CSBS

Innovative Practice Description Innovative Practice Description

Unit / Topic: Unit III / Linear Search, Binary Search

Course Outcome: CO 3

Topic Learning Outcome: TLO 10

Activity Chosen: Think Pair Share

Justification:

- Students are asked to think individually and to share the knowledge with classmates. In python, Searching is an important concept and it is necessary for students to understand the searching concept and its execution. The students' individual understanding capability can be tested and the gained ideas can be shared by think-pair-share activity.
- **Time Allotted for the Activity: 30 minutes**

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes. Instructor gave a program to the students.
- Students think about what they know or have learned in searching, and come up with their own idea. [Takes 1-5 Minutes].
- The instructor asks them to pair with nearby students to discuss the program solution. The students interact with each other about their programs to deepen understanding of the iterative.
- They share their thoughts with each other and proceed with the task. [Takes 6-10 Minutes].
- Students share their solution with the entire class [Takes 11-15 Minutes]. Instructor moderates the discussion and highlights important points.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO10	PSO3
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CO 3	2	1	1	1	1	1
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(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO10	PSO3
	2	1	1	1
Justification for correlation	Students can able to solve the various decision making and searching problems by using basics mathematics, science and engineering principles	Students can able to solve searching related problems and able to provide more accurate solutions for the complex engineering problems	Students could be able to explore their knowledge of solving the problems using branching, looping and built-in functions and searching techniques and its types by giving presentation and through discussion forum	Students can be able to create and demonstrate simple python program to provide the decision in civil engineering problem

- **Images / Screenshot of the practice:**



def linear_search(arr, x):
 for i in range(0, len(arr)):
 if arr[i] == x:
 return i
 return -1
 n = int(input("Enter the number of elements"))
 arr = []
 print("Enter the elements")
 for i in range(0, n):
 arr.append(input())
 print("The elements in the list are:", arr)
 x = input("Enter the element to be searched")
 res = linear_search(arr, x)
 if (res != -1):
 print("The element is present in the list at position", res)
 else:
 print("The element is not present")

O/P

Element found at index : 3

Binary Search

Item to be searched = 23

Step-1

4	5	7	9	12	17	22	23	28
0	1	2	3	4	5	6	7	8

$$a[mid] = 12$$

$$12 < 23$$

$$b = mid + 1$$

$$= 4 + 1$$

$$mid = \frac{(b + end)}{2} = \frac{5 + 8}{2} = \frac{13}{2} = 6$$

end = 8
mid = 4

Take the element in 6th position

Step 2:

4	5	7	9	12	17	22	23	28
0	1	2	3	4	5	6	7	8

$$a[mid] = 23$$



Fig:1 Students discussed among themselves and took individual points

Fig:2 Students shared their ideas to their classmates as a team.

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

The students felt that this activity helps them to remember the concepts and it will be very helpful to correct their mistakes while sharing the program solution to the other students.

- Benefit of the practice:**

- From this activity, the students can get more clarity in the particular topic.
 - Students can able to explain the concepts without any confusion.
 - This practice has made the class more interactive than the conventional practice

- Challenges faced in implementation:**

- All the students were unable to complete the task within allotted time
 - Not able to make all the students to involve in this activity
 - Few students found it difficult to solve the program individually. After discussion, they felt easy.

- References:**

1. <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>
2. <https://www.readingrockets.org/strategies/think-pair-share>
3. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>

Signature of Faculty Member

HOD